MILITARY SPECIFICATION

CALIBRATOR, RADIAC

1. SCOPE

1.1 This specifiation covers the design of a device to calibrate radiac survey equipment and dosimeters aboard Naval ships and at shore facilities.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood, Sheathed, Nailed and Bolted (30,000# Max.).
MIL-P-116 - Preservation, Methods of.
MIL-P-15024 - Plates, Identification—Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.
MIL-M-19590 - Marking of Commodities and Containers to Indicate Radioactive Material.
MIL-R-22732 - Reliability Requirements for Shipboard and Ground Electronic Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF COMMERCE

Code of Federal Regulations, Atomic Energy, Title 10, Parts 20 and 30
AEC Handbook of Federal Regulations Applying to Transportation of Radioactive Materials

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20360.)

FSC 6665

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:46Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Preproduction sample.- Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.3).

3.2 Material.- Material shall be in accordance with MIL-E-16400, except as otherwise specified herein. Particular attention shall be paid to the acceptable and unacceptable material requirements of MIL-E-16400.

3.2.1 The radiation shielding shall consist of an alloy of 95 percent virgin lead and 5 percent manganese.

3.2.2 Mercury shall be triple distilled laboratory grade.

3.3 General requirements.- The equipment shall meet the requirements of MIL-E-16400, except as otherwise specified herein. The design objectives, as specified in MIL-E-16400 for reliability, simplicity, ease of installation and maintenance consistent with the needs of the Naval service, shall be considered during all phases of the equipment design.

3.3.1 Code of Federal Regulations.- The equipment will contain a high intensity radiation source. It is imperative that safety of operation be considered of prime importance in the manufacture of the equipment. The radioactive source and associated shielding and manipulating gear shall conform to all AEC regulations. The contractor is required to obtain the necessary AEC license for the storage and handling of the radioactive material.

3.3.2 Source design.- The source design shall be as shown on figure 1 with no more than 3 percent Cs.134 permissible. The source activity should be approximately 30 curies. The true activity shall be of such magnitude that the full range of exposure rates from 0.25mR/hr. to at least 350mR/hr. is obtainable at the 3-1/2 inch point within the radiation chamber. The requirements of 3.4.2 and 4.5.5 shall also be satisfied.

3.4 General description.- The radiac calibrator set shall be suitable to calibrate existing Navy gamma radiac survey equipment such as:

AN/PDR-27 series	IM-9/PD series	IM-135/PD
AN/PDR-18 series	IM-107/PD	IM-181/PD
AN/PDR-45 series	IM-19, 20/PD	
AN/PDR-43 series	IM-94/PD	

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:46Z
Check the source to verify that this is the current version before use.

3.4.1 Construction.- The construction and assembly shall follow the form factor specified herein and in manufacturing drawings specified (see 6.1). Where the requirements of this specification conflict with the drawings, this specification shall govern. Any recommended design changes shall be described in detail and submitted to the bureau or agency concerned for approval. The construction of the equipment shall be such that the operator can perform all necessary manipulations easily.

3.4.2 Exposure dose rate.- The radiation exposure rate shall be no more than 2mR/hr. at any point on the external surface of the equipment and shall be no more than 0.5mR/hr. at one foot from any point on the surface for any setting of the source position.

3.4.3 Safety requirements.- Safety requirements such as external exposure rates, leak tests, Bureau of Explosives Permits and AEC licenses shall be the responsibility of the contractor. The source procurement and loading shall also be the responsibility of the contractor.

3.5 Definitions.- For purposes of this specification, the definitions specified in 6.3 shall apply.

3.6 Mechanical design.- The calibrator shall contain all parts and components shown on the specified microfilm and any other parts and components which may not be shown but necessary to performance in accordance with this specification. The calibrator shall consist of the following mechanical assemblies:

(a) Source container, including the source.
(b) Source shield.
(c) Source positioning mechanism.
(d) Irradiation box.
(e) Mounting bench.

3.6.1 Source container.- The source container shall consist of all parts necessary to hold the source of radiation in an approved manner.

3.6.1.1 Loading.- The source shall be loaded, eyelet end in, into the source container under prescribed safety conditions. The cap nut shall be torqued down on the commercially pure nickel washer to at least 40 foot pounds, or higher if necessary to assure cold-weld of the nickel surfaces to the adjoining surfaces of the source container. See referenced manufacturing drawings specified (see 6.1) for parts delineation.

3.6.2 Source shield.- Radiation from the source shall be contained primarily within a cylindrical lead shield terminated by a flat surface at one end and by a hemispherical surface at the other end. A hole shall be machined along the logitudinal axis of the cylinder and lined with a corrosion resistant steel

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:46Z
Check the source to verify that this is the current version before use.

tube called the mercury tube. This tube shall be suitably terminated at both ends to allow passage of the source positioning mechanism and retain the source container and the mercury used as the attenuating medium.

3.6.3 Source positioning mechanism.- The source positioning indicator shall be capable of reproducibly positioning the source at any predetermined point along its path of travel within its operating limits.

3.6.3.1 The continuous positioning mechanism shall drive the radiation source in a continuous manner along the longitudinal axis of the source shield, through the mercury attenuating medium. The reproducibility of any position of the source along the axis of travel shall be better than 3/1000 of an inch.

3.6.3.2 A source position indicator shall be provided which will indicate at all times the exact position of the source within the source shield. This indicator shall indicate the source position to within the reproducibility requirements specified herein and shall be placed in such a position as to be easily readable from the operator's position.

3.6.4 Irradiation box.- A safety shielded irradiation box shall be provided into which the radiac instruments to be calibrated can be placed. The box shall be provided with the following accessories:

(a) Vertical and horizontal instrument positioning devices.
(b) Movable top lid.
(c) Safety interlock mechanism.

3.6.4.1 The vertical positioning device shall be designed so that its elevation is continuously variable. This device shall be provided with a readout which will accurately indicate the elevated position of the table. The horizontal positioning device shall be in the form of an adjustable partition which can be varied in increments not exceeding 0.5 inches in a plane perpendicular to the source travel axis. The design shall be such as to permit adjustment of the horizontal positioning device independently of the vertical position device. The reproducibility of position of these positioning devices shall be better than 1/100 of an inch.

3.6.4.2 The irradiation box shall be provided with a movable top lid into which is fitted a HI-D glass window for viewing the radiac equipment within the irradiation box. This lid shall be provided with the necessary rollers and rails to slide along a horizontal plane and permit insertion of the radiac equipment in the box. The design of the top lid fittings shall be such that the lid will be safely retained under shipboard shock conditions.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:46Z
Check the source to verify that this is the current version before use.

3.6.4.3 An automatic safety interlock mechanism shall be provided so that the lid cannot be opened unless the radioactive source is first pushed back into the source holder to a "safe position". Extreme care shall be taken in the manufacture of this safety mechanism to eliminate possibility of failure. In addition to the safety mechanism, a suitable warning plate shall be affixed to the irradiation box or lid to warn the operator who is about to open the lid to make certain the source is retracted before opening the lid.

3.6.5 Mounting bench.- The source holder, irradiation box, positioning devices, and all other accessories shall be securely mounted upon a metal bench. This bench shall be adequate to support the weight of the equipment and of such height as to allow convenient operation of all controls by one man in a standing position. Means shall be provided to bolt this bench securely to a floor or deck.

3.7 Operating side.- All controls, cranks, knobs, readouts, signal lights, signs, labels and any other features of importance to the operator, shall be conveniently operable or visible from one side by one man in a normal standing position.

3.8 Size and weight.- The overall dimensions of the completely assembled calibrator shall not exceed 43 inches in length by 21 inches wide by 47 inches high.

3.8.1 The weight of the completely assembled calibrator shall not exceed 2025 pounds.

3.9 Electrical design.-

3.9.1 Power source.- The electrical components of the calibrator shall be capable of operating on an electrical energy source of 115 volts, ±10 percent, single phase, 60 cycles per second, ±5 percent. All wiring shall be of the two wire type and shall be installed to afford maximum protection against electrical shock to operating personnel. The electrical system shall be capable of withstanding 150-volts without impairment of normal functions.

3.9.2 Electrical controls.- An electrical control system, including two lamps (one green, one red) and all necessary switches, contacts, wiring, and sockets shall be provided in accordance with the applicable portion of MIL-E-16400 to perform the following functions:

(a) When the source is retracted in the safe position, the green lamp shall be on and the red lamp off.
(b) When the source is removed from the safe position, the red lamp shall be on and the green lamp off.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:46Z
Check the source to verify that this is the current version before use.

3.9.3 Mercury level indicator.- A mercury level indicator light shall be provided. This light (green) shall indicate that the mercury level in the mercury tube is adequate for safe operation of the calibrator.

3.10. External finish.- Finish shall be as specified in MIL-E-16400. Finished working surfaces shall not be painted.

3.11 Identification plates, labels and markings.- Plates shall be in accordance with type A, B or C of MIL-P-15024. All parts and subassemblies shall be identified by appropriate markings. Identification plates and labels shall be made, in descending order of preference, of nickel-copper alloy, brass, corrosion resisting steel or aluminum alloy plates of sheets of a minimum thickness which will assure permanence for the application. The material selected shall also be electrolytically compatible in a sea environment with the material to which affixed. The size and type of printing and markings shall be suitable for the purpose intended and shall be legibly indented into the surface by etching, engraving, stamping, embossing or similar method. The letters shall be filled with a contrasting, permanent material which will facilitate readability and legibility. Finished identification plates and labels shall be covered with a protective coating of clear varnish, lacquer or similar material. Plates and labels shall be positively and permanently affixed to the calibrator using screws or rivets of a material, compatible in a sea environment with the plate or label and part on which mounted. Plating or other means acceptable to the procuring activity shall be utilized, if required, to produce corrosion resistance.

3.11.1 A calibrator identification plate shall be provided. Among other necessary contractor, Government and contract data, the plate shall include the nominal strength of the source and its date of determination as furnished by the laboratory which provides the source; a serial number shall also be included which will be furnished, upon written request to the bureau or agency concerned.

3.11.2 A radiation warning shall be placed in an appropriate location on the exterior of each source shield and where it will be readily visible. This label shall bear the conventional radiation symbol (magenta on yellow background), with marking in accordance with MIL-M-19590.

3.11.3 A label bearing operating instructions shall be placed on the exterior of the equipment where it will be readily visible.

3.11.4 Markings of parts and subassemblies shall be legible and made by a method which will produce markings of sufficient durability to last the anticipated life of the part or subassembly under the environmental conditions to which it will be subjected. Care shall be taken not to damage or affect the utility of parts or subassemblies when marking by such methods as stamping or similar means.

Source: https://assist.dla.mil -- Downloaded: 20260-19T18:46Z
Check the source to verify that this is the current version before use.

3.12 Shock and vibration.- The calibrator shall be designed in accordance with the criteria for shock and vibration set forth in MIL-E-16400.

3.13 Reliability.- The following procedure outlines the steps necessary to fulfill the requirements specified in MIL-E-16400 for reliability as it applies to this equipment.

3.13.1 Level.- The reliability requirement for this equipment (MTBF) is not applicable.

3.13.2 Reliability evaluation shall be performed in accordance with the procedures for "Reliability assurance by analysis and prediction" as specified in MIL-R-22732.

3.13.3 In addition to the regular reliability reports of MIL-R-22732, reliability prediction reports shall be submitted whenever the design changes or additional design data becomes available affecting the prediction.

3.14 Workmanship.- Workmanship shall be as specified in MIL-E-16400. Special attention shall be given to the workmanship which enters into the fabrication of parts associated with shielding, supporting and positioning the radioactive source. The quality of the shielding materials shall be such as to attenuate radiation to the limits specified herein as a maximum.

3.15 Technical manuals.- Two manuals shall be supplied with each unit and shall include all information necessary for installation, operation and maintenance of the equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General inspection.- The general inspection of the equipment shall be comprised of the following classifications as specified in MIL-E-16400 and as modified herein:

(a) Preproduction inspection.
(b) Production inspection.

Source: https://assist.dla.mil -- Downloaded: 20260-19T18:46Z
Check the source to verify that this is the current version before use.

4.2.1 Proposed tests and test procedures.- A list of proposed tests, test procedures and test equipment shall be furnished by the contractor to the Government inspector for transmittal to the bureau or agency concerned for approval 60 days prior to setting up test equipment for preproduction or quality conformance inspection.

4.3 Preproduction inspection.- Preproduction inspection shall consist of the examination and tests specified in 4.3.1.

4.3.1 The preproduction model shall be tested as specified in 4.5 and the following tests of MIL-E-16400:

(a) Surface examination.
(b) Operating test.
(c) Weights and dimensions.
(d) Supply line voltage and frequency.
(e) Shock and vibration.

4.3.1.1 Shock and vibration tests.- The equipment shall be shock tested as specified for grade B, class I, medium weight, type A. Shock and vibration tests shall be conducted prior to loading, but with a dummy source in position in the mercury. After test, the mechanical reproducibility of the source positioning mechanism shall be unimpaired.

4.3.1.1.1 Disposition of shock tested equipment.- A shock tested unit shall be thoroughly reconditioned by the contractor before resubmitting it as production equipment.

4.4 Production inspection.- Production inspection shall be as specified in MIL-E-16400 and in addition shall include the tests of 4.5.

4.5. Inspection.-

4.5.1 Controls and mechanisms.- All controls and mechanisms shall be inspected for satisfactory operation and conformance with the applicable portions of this specification before and after loading.

4.5.2 Pre-loading tests.- Tests shall be conducted in accordance with Title 10 of the Code of Federal Regulations on the sources immediately upon receipt to determine that they have not been damaged in transit or are otherwise leaking or contaminated. If any of the tests reveal the presence of radioactivity in excess of allowable limits which cannot be removed or reduced by prescribed methods, the source shall not be loaded into a calibrator source container.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:46Z
Check the source to verify that this is the current version before use.

4.5.3 Background tests.- After assembling the calibrator and setting the travel limits of the source, the equipment shall be tested to determine that the exposure dose rates specified in 3.4.2 are not exceeded. If the exposure dose rates are exceeded, drawings showing the proposed additional shielding required to achieve the required rates shall be forwarded for approval to the bureau or agency concerned.

4.5.4 Periodic tests.- Periodic tests required by Title 10 of the Code of Federal Regulations shall be conducted.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Source drive screws, gears and unpainted rails shall be cleaned and coated with preservative type P-2 of MIL-P-116 for protection against corrosion. All components shall be adequately secured in position to prevent movement in transit. Each radiac calibrator shall be packed in a fully sheathed crate conforming to MIL-C-104. Crates shall have sides, top and ends bolted to the base in a manner to permit removal as a "shell" in order to facilitate disassembly of the crate, removal of the calibrator from the container and repacking in the same container. Thru bolts (such as United Car Fasteners Corporation's "tee-nut" fasteners) shall be used in lieu of lag bolts since the holding power of lag bolts is considerably lessened after a few openings and closures. The calibrator shall be adequately secured to the base and blocked and braced to prevent damage in transit. All floor boards within the fork-lift truck finger areas shall be bolted or otherwise blocked to the skids to prevent the floor boards from lifting out. Construction of the container shall comply in all other respects to the AEC Handbook of Federal Regulations Applying to Transportation of Radioactive Materials.

5.2 Marking.- Shipments shall be marked in accordance with MIL-M-19590 with the following identification information:

Stock number
Nomenclature
Unit (Quantity)
Serial number
Gross weight
Cube

5.2.1 Contract data marking shall consist of:

Requisition or project number
Inspection stamp

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:46Z
Check the source to verify that this is the current version before use.

5.2.2 After the radiation source has been inserted in the container, containers shall also be marked in accordance with the AEC Handbook of Federal Regulations Applying to Transportation of Radioactive Materials.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Manufacturing drawings (see 3.4.1 and 3.6.1.1).

6.2 Transportation permit.- Bureau of Explosives has assigned Permit Number 876 to the transportation of loaded radiac calibrators of this design. Correspondence relative to each shipment of calibrators should refer to this permit number and be directed to Bureau of Explosives, Association of American Railroads, 63 Vesey Street, New York City, New York, Zip 10007.

6.3 Definitions.- Definitions shall be as follows:

6.3.1 Source.- The source is the assembly consisting of the CS-137 isotopic material and its sealed corrosion resisting steel enclosure, as shown on figure 1.

6.3.2 Source container.- The enclosure at the end of and which is an integral part of the source positioning rod. The source fits into the source container.

6.3.3 Mercury tube.- The mercury tube is the one-inch diameter horizontal corrosion resistant steel tube which contains the mercury bath, the source container and positioning rod, and the source.

6.3.4 Source shield.- The source shield is the lead-filled belt-shaped casting which holds the mercury tube and its contents, including the source.

6.4 Preproduction.-

6.4.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior to Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - Ships
(Project 6665-N157Sh)

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:46Z
Check the source to verify that this is the current version before use.

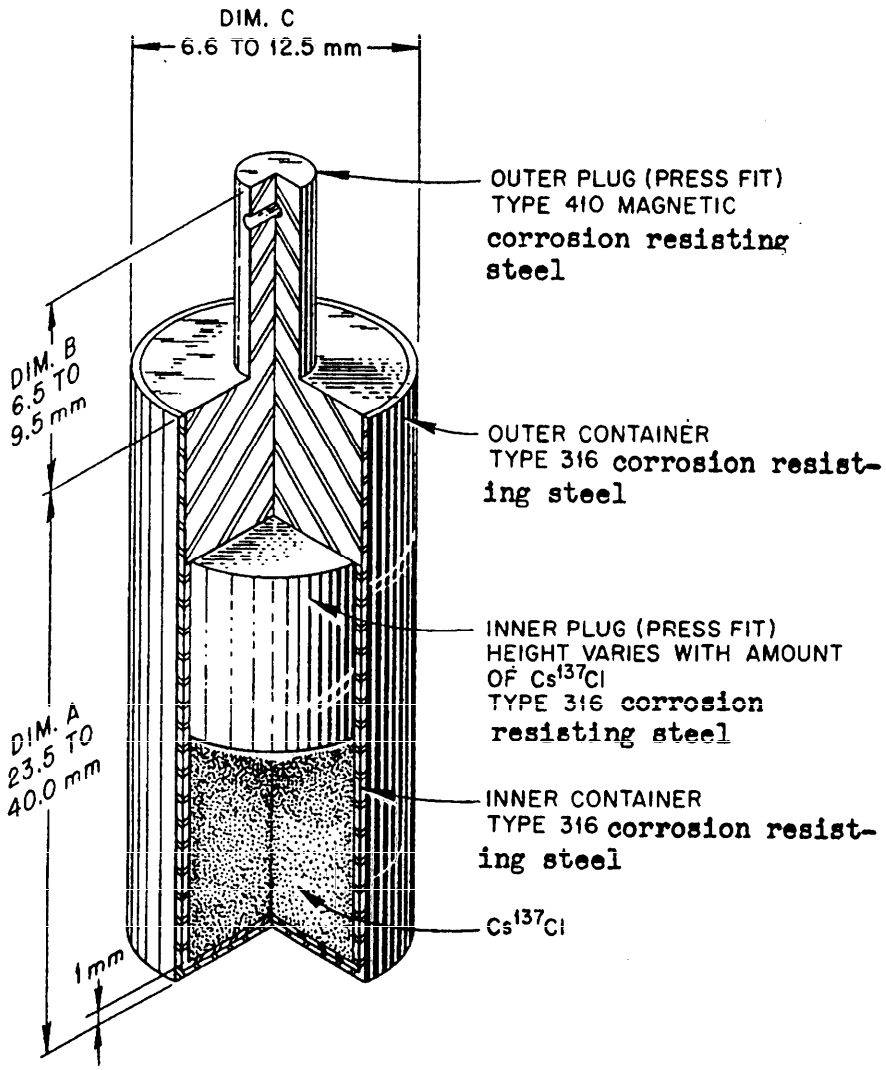


Fig. 1 – Pelleted Cs^{137} Gamma Source, 5 to 130 Curies of Activity.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T18:46Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:46Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A.

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

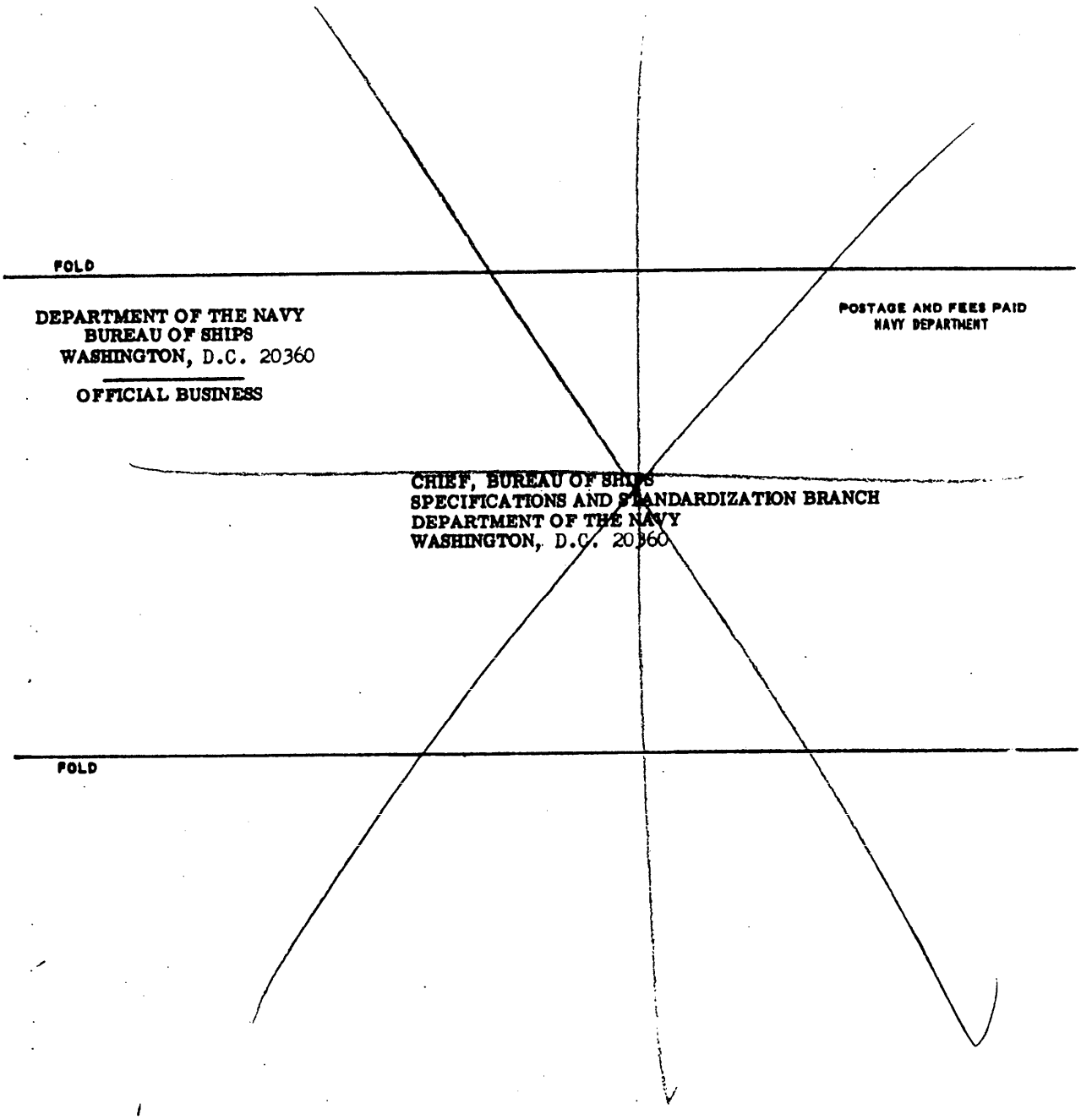
4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

BuSHIPS 700

Source: https://assist.dla.mil -- Downloaded: 202640-19T18:46Z
Check the source to verify that this is the current version before use.



FOLD

DEPARTMENT OF THE NAVY
BUREAU OF SHIPS
WASHINGTON, D.C. 20360

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
NAVY DEPARTMENT

CHIEF, BUREAU OF SHIPS
SPECIFICATIONS AND STANDARDIZATION BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 20261O-19T18:46Z
Check the source to verify that this is the current version before use.